

矩阵分析教程

(电子版)

董增福

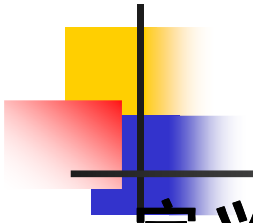
哈尔滨工业大学数学系



第八章 矩阵的Kronecker积及其应用

核心内容：

1. 矩阵的 Kronecker 积
2. 矩阵Kronecker 积的特征值
3. 求解矩阵方程
4. 求解矩阵微分方程



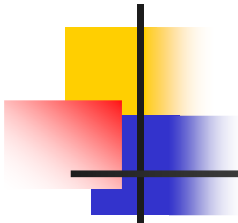
§ 8.1 矩阵的Kronecker积

定义 8.1 设 $A = (a_{ij})_{m \times n} \in \mathbb{C}^{m \times n}$, $B = (b_{st})_{p \times q} \in \mathbb{C}^{p \times q}$,

称如下的分块矩阵

$$A \otimes B \triangleq \begin{bmatrix} a_{11}B & \cdots & a_{1j}B & \cdots & a_{1n}B \\ \vdots & & \vdots & & \vdots \\ a_{i1}B & \cdots & a_{ij}B & \cdots & a_{in}B \\ \vdots & & \vdots & & \vdots \\ a_{m1}B & \cdots & a_{mj}B & \cdots & a_{mn}B \end{bmatrix} = (a_{ij}B) \in \mathbb{C}^{mp \times nq} \quad (8.1)$$

为 A 与 B 的Kronecker积, 也称为矩阵 A 与 B 的直积或张量积.



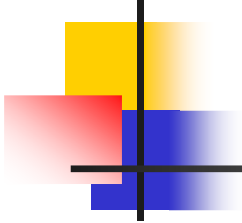
由上述定义有

$$\mathbf{B} \otimes \mathbf{A} = (b_{st} \mathbf{A}) \in \mathbf{C}^{pm \times qn} \quad (8.2)$$

显然 $\mathbf{A} \otimes \mathbf{B}$ 与 $\mathbf{B} \otimes \mathbf{A}$ 是同型矩阵,一般地

$$\mathbf{A} \otimes \mathbf{B} \neq \mathbf{B} \otimes \mathbf{A}$$

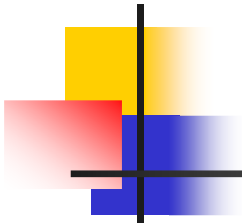
即矩阵的Kronecker积不满足交换律.



例 8.1 设 $A = \begin{bmatrix} 1 & 4 \\ 0 & 2 \end{bmatrix}$, $B = \begin{bmatrix} 3 & -1 \end{bmatrix}$, 则

$$A \otimes B = \begin{bmatrix} 1B & 4B \\ 0B & 2B \end{bmatrix} = \begin{bmatrix} \boxed{3} & \boxed{-1} & \boxed{12} & \boxed{-4} \\ \boxed{0} & \boxed{0} & \boxed{6} & \boxed{-2} \end{bmatrix},$$

$$B \otimes A = \begin{bmatrix} 3A & -1A \end{bmatrix} = \begin{bmatrix} \boxed{3} & \boxed{12} & \boxed{-1} & \boxed{-4} \\ \boxed{0} & \boxed{6} & \boxed{0} & \boxed{-2} \end{bmatrix}.$$



显然 $A \otimes B \neq B \otimes A$, 即矩阵的
Kronecker 积不满足交换律.

但是对于单位矩阵 I_m, I_n 有

$$I_m \otimes I_n = I_n \otimes I_m = I_{mn}. \quad (8.3)$$



定理 8.1 矩阵的 **Kronecker 积** 有如下的 **基本性质**:

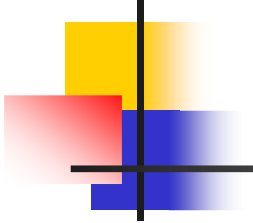
(1) 设 k 为复常数, 则

$$(kA) \otimes B = A \otimes (kB) = k(A \otimes B); \quad \text{齐次性(8.4)}$$

$$(2) (A \otimes B) \otimes C = A \otimes (B \otimes C); \quad \text{结合律(8.5)}$$

$$(3) (A_1 + A_2) \otimes B = A_1 \otimes B + A_2 \otimes B, \quad \text{分配律(8.6a)}$$

$$B \otimes (A_1 + A_2) = B \otimes A_1 + B \otimes A_2; \quad (8.6b)$$



$$(4) \quad (A \otimes B)^T = A^T \otimes B^T \quad (8.7)$$

$$(A \otimes B)^H = A^H \otimes B^H \quad (8.8)$$

(5) 设 $A = (a_{ij})_{m \times n}$, $B = (b_{ij})_{p \times q}$,

$C = (c_{ij})_{n \times s}$, $D = (d_{ij})_{q \times t}$, 则

$$(A \otimes B)(C \otimes D) = (AC) \otimes (BD) \quad (8.9)$$

特别当 $A \in \mathbb{C}^{m \times m}$, $B \in \mathbb{C}^{n \times n}$ 时,

$$(A \otimes B)^k = A^k \otimes B^k$$

(6) 设 $A \in \mathbb{C}^{m \times m}$, $B \in \mathbb{C}^{n \times n}$ 都是可逆矩阵, 则 $A \otimes B$

也可逆, 且有

$$(A \otimes B)^{-1} = A^{-1} \otimes B^{-1} \quad (8.10)$$

$$(7) \quad (A \otimes B)^+ = A^+ \otimes B^+ \quad (8.11)$$

$$(8) \text{ 设 } A \in U^{m \times m}, B \in U^{n \times n}, \text{ 则 } A \otimes B \in U^{mn \times mn} \quad (8.12)$$

$$(9) \text{ 设 } A \in \mathbb{C}^{m \times m}, B \in \mathbb{C}^{n \times n}, \text{ 则 } A \otimes B \sim B \otimes A$$

证 由定义 8.1 即可证出 (1) ~ (4)

例如 (2) 表示满足结合律, 证明如下:

$$\begin{aligned} (A \otimes B) \otimes C &= \begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & & \vdots \\ a_{m1} & \cdots & a_{mn} \end{bmatrix} \otimes B \otimes C = \begin{bmatrix} a_{11}B & \cdots & a_{1n}B \\ \vdots & & \vdots \\ a_{m1}B & \cdots & a_{mn}B \end{bmatrix} \otimes C \\ &= \begin{bmatrix} (a_{11}B) \otimes C & \cdots & (a_{1n}B) \otimes C \\ \vdots & & \vdots \\ (a_{m1}B) \otimes C & \cdots & (a_{mn}B) \otimes C \end{bmatrix} = \begin{bmatrix} a_{11}(B \otimes C) & \cdots & a_{1n}(B \otimes C) \\ \vdots & & \vdots \\ a_{m1}(B \otimes C) & \cdots & a_{mn}(B \otimes C) \end{bmatrix} \\ &= A \otimes (B \otimes C) \end{aligned}$$

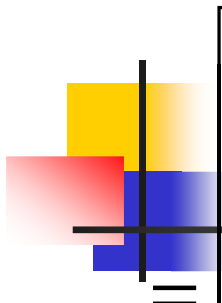
(4) 中(8.8)式 $(A \otimes B)^H = A^H \otimes B^H$ 证明如下:

$$\begin{aligned}
 (A \otimes B)^H &= \begin{bmatrix} a_{11}B & \cdots & a_{1n}B \\ \vdots & & \vdots \\ a_{m1}B & \cdots & a_{mn}B \end{bmatrix}^H = \begin{bmatrix} \overline{a_{11}}B^H & \cdots & \overline{a_{m1}}B^H \\ \vdots & & \vdots \\ \overline{a_{1n}}B^H & \cdots & \overline{a_{mn}}B^H \end{bmatrix} \\
 &= \begin{bmatrix} \overline{a_{11}} & \cdots & \overline{a_{m1}} \\ \vdots & & \vdots \\ \overline{a_{1n}} & \cdots & \overline{a_{mn}} \end{bmatrix} \otimes B^H = A^H \otimes B^H
 \end{aligned}$$

可见(8.8)式成立, 将 H 换成 T , 则(8.7)式成立. 11

(5). 由定义 8.1 可得

$$\begin{aligned} (A \otimes B)(C \otimes D) &= \begin{bmatrix} a_{11}B & \cdots & a_{1n}B \\ \vdots & & \vdots \\ a_{m1}B & \cdots & a_{mn}B \end{bmatrix} \begin{bmatrix} c_{11}D & \cdots & c_{1s}D \\ \vdots & & \vdots \\ a_{n1}D & \cdots & a_{ns}D \end{bmatrix} \\ &= \begin{bmatrix} \sum_{k=1}^n (a_{1k}B)(c_{k1}D) & \cdots & \sum_{k=1}^n (a_{1k}B)(c_{ks}D) \\ \vdots & & \vdots \\ \sum_{k=1}^n (a_{mk}B)(c_{k1}D) & \cdots & \sum_{k=1}^n (a_{mk}B)(c_{ks}D) \end{bmatrix} \end{aligned}$$



$$\begin{aligned}
 &= \begin{bmatrix} \sum_{k=1}^n a_{1k} c_{k1} (\textcolor{red}{B}\textcolor{violet}{D}) & \cdots & \sum_{k=1}^n a_{1k} c_{ks} (\textcolor{red}{B}\textcolor{violet}{D}) \\ \vdots & & \vdots \\ \sum_{k=1}^n a_{mk} c_{k1} (\textcolor{red}{B}\textcolor{violet}{D}) & \cdots & \sum_{k=1}^n a_{mk} c_{ks} (\textcolor{red}{B}\textcolor{violet}{D}) \end{bmatrix} \\
 &= \begin{bmatrix} \sum_{k=1}^n a_{1k} c_{k1} & \cdots & \sum_{k=1}^n a_{1k} c_{ks} \\ \vdots & & \vdots \\ \sum_{k=1}^n a_{mk} c_{k1} & \cdots & \sum_{k=1}^n a_{mk} c_{ks} \end{bmatrix} \otimes (\textcolor{red}{B}\textcolor{violet}{D}) = (\textcolor{blue}{A}\textcolor{black}{C}) \otimes (\textcolor{red}{B}\textcolor{violet}{D})
 \end{aligned}$$

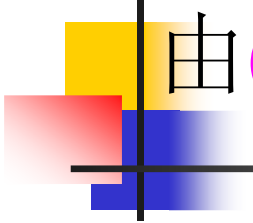
(6).由(5)有

$$(A \otimes B)(A^{-1} \otimes B^{-1}) = (AA^{-1}) \otimes (BB^{-1}) = I_m \otimes I_n = I_{mn}.$$

由 $A \otimes B$ 与 $A^{-1} \otimes B^{-1}$ 都是 mn 阶的方阵, 这说明矩阵 $A \otimes B$ 可逆, 并且它的逆矩阵就是 $A^{-1} \otimes B^{-1}$, 即

$$(A \otimes B)^{-1} = A^{-1} \otimes B^{-1}.$$

(7). 这只要验证 **Penrose** 方程中的四个等式即可 (其中用 $A \otimes B$ 代替 A , 用 $A^+ \otimes B^+$ 代替 B), 验证如下:



由(5)验证第①个, 第②个 **Penrose** 方程

$$\begin{aligned} & (A \otimes B)(A^+ \otimes B^+)(A \otimes B) \\ &= (AA^+A) \otimes (BB^+B) = A \otimes B. \end{aligned} \quad \text{①成立;}$$

$$\begin{aligned} & (A^+ \otimes B^+)(A \otimes B)(A^+ \otimes B^+) \\ &= (A^+AA^+) \otimes (B^+BB^+) = A^+ \otimes B^+ \end{aligned} \quad \text{②成立;}$$

由(5)和(4)验证第③个,第④个 **Penrose** 方程

$$\begin{aligned} [(A \otimes B)(A^+ \otimes B^+)]^H &= [(AA^+) \otimes (BB^+)]^H = (AA^+)^H \otimes (BB^+)^H \\ &= (AA^+) \otimes (BB^+) = (A \otimes B)(A^+ \otimes B^+) \quad \text{③成立;} \end{aligned}$$

$$\begin{aligned} [(A^+ \otimes B^+)(A \otimes B)]^H &= [(A^+A) \otimes (B^+B)]^H = (A^+A)^H \otimes (B^+B)^H \\ &= (A^+A) \otimes (B^+B) = (A^+ \otimes B^+)(A \otimes B) \quad \text{④成立} \end{aligned}$$

这说明 $A^+ \otimes B^+$ 是 $A \otimes B$ 的加号逆, 由其惟一性, 即得

$$(A \otimes B)^+ = A^+ \otimes B^+.$$

(8). 设 $A \in U^{m \times m}$, $B \in U^{n \times n}$, 由(4)及(5)得

$$\begin{aligned}(A \otimes B)(A \otimes B)^H &= (A \otimes B)(A^H \otimes B^H) \\ &= (AA^H)(BB^H) = I_m \otimes I_n = I_{mn}.\end{aligned}$$

$$\begin{aligned}(A \otimes B)^H (A \otimes B) &= (A^H \otimes B^H)(A \otimes B) \\ &= (A^H A)(B^H B) = I_m \otimes I_n = I_{mn}.\end{aligned}$$

故 $(A \otimes B)(A \otimes B)^H = (A \otimes B)^H (A \otimes B) = I_{mn},$

即 $A \otimes B \in U^{mn \times mn}.$

(9). 参考《矩阵分析教程》P291.16题.

Kronecker积的特性

除了具有齐次性,结合律,分配律之外,
矩阵的Kronecker乘积运算还可以概括出如下特性:
(假设满足运算合理性所需的条件)

① 前前后后性 $(A \otimes B)(C \otimes D) = (AC) \otimes (BD)$

② 乘幂算子性 $(A \otimes B)^* = A^* \otimes B^*$

这里 $*$ 可以代表 $T, H, -1, k, +$ 等运算

③ 酉不变性 两个酉矩阵的Kronecker乘积仍是酉矩阵.

§ 8.2 矩阵Kronecker积的特征值

定理 8.2 设 $A \in C^{m \times m}$ 的特征值为 $\lambda_1, \dots, \lambda_i, \dots, \lambda_m$,
 $B \in C^{n \times n}$ 的特征值为 $\mu_1, \dots, \mu_j, \dots, \mu_n$, 则

(1) $A \otimes B$ 的 mn 个特征值为 $\lambda_i \mu_j$

$$(i = 1, 2, \dots, m; j = 1, 2, \dots, n); \quad (8.13)$$

(2) $A \otimes I_n + I_m \otimes B^T$ 的 mn 个特征值为 $\lambda_i + \mu_j$

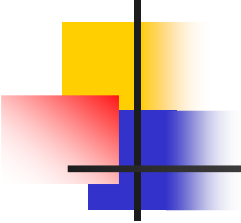
$$(i = 1, 2, \dots, m; j = 1, 2, \dots, n). \quad (8.14)$$

(2) 式中为 B^T 而非 B , 是为后用.

证 用**Jordan 标准形**来完成本定理的证明.

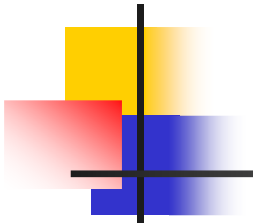
(1). 由**定理 3.10**, 对于矩阵 **A** 与 **B** , 存在**可逆**
矩阵 **P_1** , **P_2** , 使

$$P_1^{-1}AP_1 = \begin{bmatrix} \lambda_1 & \varepsilon_1 & & & \\ & \lambda_2 & \ddots & & \\ & & \ddots & \varepsilon_{i-1} & \\ & & & \lambda_i & \ddots \\ & & & & \ddots & \varepsilon_{m-1} \\ & & & & & \lambda_m \end{bmatrix} = J_1,$$



$$P_2^{-1}BP_2 = \begin{bmatrix} \mu_1 & \delta_1 & & & \\ & \mu_2 & \ddots & & \\ & & \ddots & \delta_{j-1} & \\ & & & \mu_j & \ddots \\ & & & & \ddots & \delta_{n-1} \\ & & & & & \mu_n \end{bmatrix} = J_2$$

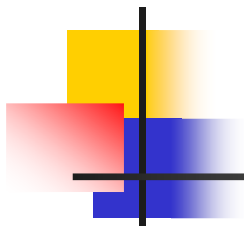
这里 ε_i, δ_j 为1或0, 其中 $i = 1, \dots, m-1; j = 1, \dots, n-1$.



由**定理** 8.1之(5)与(6)有

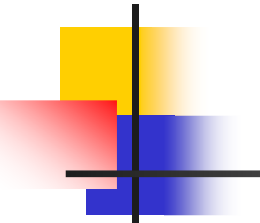
$$\begin{aligned} & (P_1 \otimes P_2)^{-1} (A \otimes B) (P_1 \otimes P_2) \\ &= (P_1^{-1} \otimes P_2^{-1}) (A \otimes B) (P_1 \otimes P_2) \\ &= (P_1^{-1} A P_1) \otimes (P_2^{-1} B P_2) \\ &= J_1 \otimes J_2. \end{aligned}$$

这说明 $A \otimes B$ 与 $J_1 \otimes J_2$ 相似，
故它们的特征值相等。



$$\boxed{J_1 \otimes J_2} = \begin{bmatrix}
 \lambda_1 J_2 & \varepsilon_1 J_2 & & \\
 & \lambda_2 J_2 & & \\
 & & \ddots & \\
 & & & \varepsilon_{i-1} J_2 \\
 & & & \boxed{\lambda_i J_2} \\
 & & & & \ddots \\
 & & & & & \varepsilon_{m-1} J_2 \\
 & & & & & & \lambda_m J_2
 \end{bmatrix}$$

其中



$$\boxed{\lambda_i J_2} = \begin{bmatrix} \lambda_i \mu_1 & \lambda_i \delta_1 & & \\ & \lambda_i \mu_2 & \ddots & \\ & & \ddots & \lambda_i \delta_{j-1} \\ & & & \boxed{\lambda_i \mu_j} & \ddots \\ & & & & \ddots & \lambda_i \delta_{n-1} \\ & & & & & \lambda_i \mu_n \end{bmatrix},$$

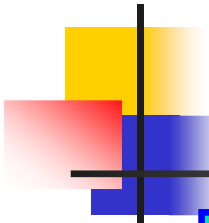
$i = 1, \dots, m.$

可见 $J_1 \otimes J_2$ 是上三角矩阵, 它的特征值恰为
对角元 $\boxed{\lambda_i \mu_j}$ ($i = 1, \dots, m; j = 1, \dots, n$).

(2). 因 $\det(\lambda I - B) = \det(\lambda I - B^T)$, 故 B^T 与 B 特征值相同,
 则存在可逆矩阵 P_3 , 化 B^T 为它的 Jordan 标准形 J_3 ,

$$P_3^{-1}(B^T)P_3 = \begin{bmatrix} \mu_1 & \sigma_1 & & & \\ & \mu_2 & \ddots & & \\ & & \ddots & \sigma_{j-1} & \\ & & & \mu_j & \ddots \\ & & & & \ddots & \sigma_{n-1} \\ & & & & & \mu_n \end{bmatrix} = J_3$$

这里 σ_j 为 1 或 0, 其中 $j = 1, \dots, n-1$.



于是有

$$\begin{aligned} & (\mathbf{P}_1 \otimes \mathbf{P}_3)^{-1} (\mathbf{A} \otimes \mathbf{I}_n + \mathbf{I}_m \otimes \mathbf{B}^T) (\mathbf{P}_1 \otimes \mathbf{P}_3) \\ &= (\mathbf{P}_1^{-1} \otimes \mathbf{P}_3^{-1}) (\mathbf{A} \otimes \mathbf{I}_n + \mathbf{I}_m \otimes \mathbf{B}^T) (\mathbf{P}_1 \otimes \mathbf{P}_3) \\ &= (\mathbf{P}_1^{-1} \mathbf{A} \mathbf{P}_1) \otimes (\mathbf{P}_3^{-1} \mathbf{I}_n \mathbf{P}_3) + (\mathbf{P}_1^{-1} \mathbf{I}_m \mathbf{P}_1) \otimes (\mathbf{P}_3^{-1} \mathbf{B}^T \mathbf{P}_3) \\ &= \mathbf{J}_1 \otimes \mathbf{I}_n + \mathbf{I}_m \otimes \mathbf{J}_3 \end{aligned}$$

这说明

$A \otimes I_n + I_m \otimes B^T$ 与 $J_1 \otimes I_n + I_m \otimes J_3$ 相似,

故它们的特征值相等.

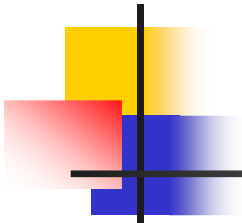
显然 $J_1 \otimes I_n, I_m \otimes J_3$ 都是上三角矩阵,

故 $J_1 \otimes I_n + I_m \otimes J_3$ 也是上三角矩阵.

而它的特征值恰为 $\lambda_i + \mu_j$

这里 $i = 1, 2, \dots, m; j = 1, 2, \dots, n$.

因此 $A \otimes I_n + I_m \otimes B^T$ 的 mn 个特征值
也为 $\lambda_i + \mu_j$. ■



定理 8.3 设 $A \in C^{m \times m}$ 的特征值为 $\lambda_1, \dots, \lambda_i, \dots, \lambda_m$,
 $B \in C^{n \times n}$ 的特征值为 $\mu_1, \dots, \mu_j, \dots, \mu_n$, 则

(1). $\det(A \otimes B) = (\det A)^n (\det B)^m$; (8.15)

(2). $\text{tr}(A \otimes B) = \text{tr}(A) \text{tr}(B)$. (8.16)

证 (1). 由定理 8.2 之 (1) 知

$$\begin{aligned}
 \det(A \otimes B) &= \prod_{i=1}^m \prod_{j=1}^n \lambda_i \mu_j = \prod_{i=1}^m \left(\lambda_i^n \prod_{j=1}^n \mu_j \right) \\
 &= \left(\prod_{i=1}^m \lambda_i^n \right) \left(\prod_{j=1}^n \mu_j \right)^m = \left(\prod_{i=1}^m \lambda_i \right)^n \left(\prod_{j=1}^n \mu_j \right)^m \\
 &= (\det A)^n (\det B)^m
 \end{aligned}$$

$$(2). \operatorname{tr}(A \otimes B) = \sum_{i=1}^m \sum_{j=1}^n \lambda_i \mu_j$$

$$= \left(\sum_{i=1}^m \lambda_i \right) \left(\sum_{j=1}^n \mu_j \right) = \operatorname{tr}(A) \operatorname{tr}(B).$$

定理 8.4 设 $A \in C_{r_1}^{m \times n}$, $B \in C_{r_2}^{p \times q}$, 则

$$\text{rank}(A \otimes B) = r_1 \cdot r_2 = \text{rank}(A) \cdot \text{rank}(B)$$

证 因为 $\text{rank}(A) = r_1$, $\text{rank}(B) = r_2$, 所以存在可逆矩阵 $P, Q; M, N$, 使

$$PAQ = \begin{bmatrix} I_{r_1} & O \\ O & O \end{bmatrix} = A_1, \quad MBN = \begin{bmatrix} I_{r_2} & O \\ O & O \end{bmatrix} = B_1, \text{ 且有}$$

$$\text{rank}(A_1) = r_1, \text{rank}(B_1) = r_2.$$

$$\text{于是} \quad A = P^{-1} A_1 Q^{-1}, \quad B = M^{-1} B_1 N^{-1}.$$

由定理 8.1之(5)有


$$\begin{aligned} A \otimes B &= (P^{-1} A_1 Q^{-1}) \otimes (M^{-1} B_1 N^{-1}) \\ &= (P^{-1} \otimes M^{-1})(A_1 \otimes B_1)(Q^{-1} \otimes N^{-1}). \end{aligned}$$

由定理 8.1之(6)知 $P^{-1} \otimes M^{-1}$, $Q^{-1} \otimes N^{-1}$ 都是可逆矩阵,

因此 $A \otimes B$ 与 $A_1 \otimes B_1$ 等价.

所以它们的秩相等, 即

$$\begin{aligned} \text{rank}(A \otimes B) &= \text{rank}(A_1 \otimes B_1) \\ &= r_1 \cdot r_2 = \text{rank}(A) \cdot \text{rank}(B). \end{aligned}$$



定理 8.5 设 x 是 $A \in C^{m \times m}$ 的特征向量, y 是 $B \in C^{n \times n}$ 的特征向量, 则 $x \otimes y$ 是 $A \otimes B$ 的特征向量.

证 设 $Ax = \lambda x$, $By = \mu y$, 由定理 8.1 之 (5) 有

$$\begin{aligned} (A \otimes B)(x \otimes y) &= (Ax) \otimes (By) \\ &= (\lambda x) \otimes (\mu y) = (\lambda \mu)(x \otimes y). \end{aligned}$$

由 $x \neq \theta$, $y \neq \theta$, 故 $(x \otimes y) \neq \theta$,

这说明 $x \otimes y$ 是矩阵 $A \otimes B$ 的属于特征值 $\lambda \mu$ 的特征向量. ■

例8.2 若 $A \in \mathbb{C}^{n \times n}$ 有 n 个线性无关的特征向量, 则 $A \otimes A$ 有 n^2 个线性无关的特征向量.

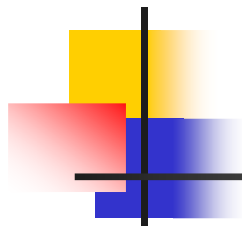
证 设 $x_1, \dots, x_i, \dots, x_n$ 为 A 的属于特征值 $\lambda_1, \dots, \lambda_i, \dots, \lambda_n$ 的 n 个线性无关的特征向量, 于是有

$$Ax_i = \lambda_i x_i \quad i=1, 2, \dots, n.$$

记 $P = (x_1, \dots, x_i, \dots, x_n)$, 则

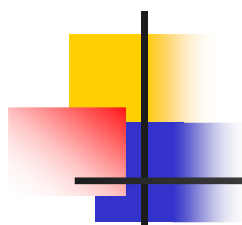
$$P^{-1}AP = \text{diag}(\lambda_1, \dots, \lambda_i, \dots, \lambda_n) = D,$$

$$\begin{aligned} & (P \otimes P)^{-1} (A \otimes A) (P \otimes P) \\ &= (P^{-1}AP) \otimes (P^{-1}AP) = D \otimes D. \end{aligned}$$



显然 $D \otimes D$ 是 n^2 阶的对角阵, 因此 $A \otimes A$ 相似于 n^2 阶的对角阵, 它有 n^2 个线性无关的特征向量.

以下证它们即为 $P \otimes P$ 的 n^2 个列向量.
由矩阵直积的定义有

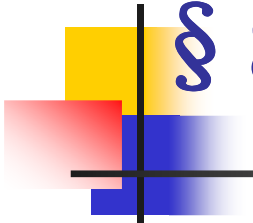


$$\begin{aligned} P \otimes P &= (x_1, \dots, x_i, \dots, x_n) \otimes P \\ &= (x_1 \otimes P, \dots, x_i \otimes P, \dots, x_n \otimes P) = \\ &= (x_1 \otimes x_1, \dots, x_1 \otimes x_i, \dots, x_1 \otimes x_n, x_2 \otimes x_1, \dots, x_i \otimes x_j, \dots, x_n \otimes x_n) \end{aligned}$$

可见 $x_i \otimes x_j$ ($i = 1, 2, \dots, n; j = 1, 2, \dots, n$.)

这 n^2 个 $P \otimes P$ 的 **列向量** 即为 $A \otimes A$ 的 n^2 个 **线性无关的特征向量** .





§ 8.3 用Kronecker积求解矩阵方程

定义 8.3 设 $A = (a_{ij})_{m \times n}$, 称 mn 维列向量

$$\text{vec } A = (a_{11}, \dots, a_{1n}, a_{21}, \dots, a_{2n}, \dots, a_{m1}, \dots, a_{mn})^T \quad (8.18)$$

为矩阵 A 的按行展开或矩阵 A 的按行拉直的列向量.

类似地可以定义矩阵 A 的按列展开或矩阵 A 的按列拉直的列向量.

例 8.3 设 $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$,

则 $\text{vec } A = (1, 2, 3, 4, 5, 6, 7, 8, 9)^T$. (列向量)

显然 $A^T = \begin{bmatrix} 1 & 4 & 7 \\ 2 & 5 & 8 \\ 3 & 6 & 9 \end{bmatrix} = (\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3)$, 则 $\text{vec } A = \begin{bmatrix} \mathbf{x}_1 \\ \mathbf{x}_2 \\ \mathbf{x}_3 \end{bmatrix}$.

此时 $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} = \begin{bmatrix} \mathbf{x}_1^T \\ \mathbf{x}_2^T \\ \mathbf{x}_3^T \end{bmatrix}$.

定理 8.7 设 $A \in C^{m \times n}$, $X \in C^{n \times p}$, $B \in C^{p \times q}$, 则

$$\text{vec}(AXB) = (A \otimes B^T) \text{vec } X \quad (8.19)$$

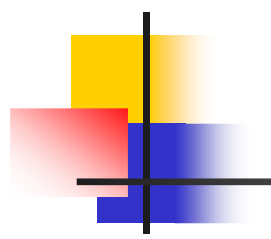
证 设 $A = (a_{ij})_{m \times n}$, $X^T = (\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n)$, $X = \begin{bmatrix} \mathbf{x}_1^T \\ \mathbf{x}_2^T \\ \vdots \\ \mathbf{x}_n^T \end{bmatrix}$, 则

$$\text{vec } X = \begin{bmatrix} \mathbf{x}_1 \\ \mathbf{x}_2 \\ \vdots \\ \mathbf{x}_n \end{bmatrix}.$$

于是

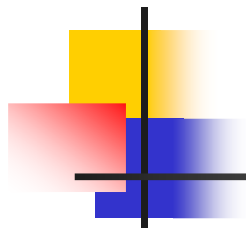
$$A \mathbf{X} B = A(\mathbf{X} B)$$

$$= \begin{bmatrix} a_{11} & \cdots & a_{1j} & \cdots & a_{1n} \\ \vdots & & \vdots & & \vdots \\ a_{i1} & \cdots & a_{ij} & \cdots & a_{in} \\ \vdots & & \vdots & & \vdots \\ a_{m1} & \cdots & a_{mj} & \cdots & a_{mn} \end{bmatrix} \begin{bmatrix} \mathbf{x}_1^T B \\ \vdots \\ \mathbf{x}_j^T B \\ \vdots \\ \mathbf{x}_n^T B \end{bmatrix}$$

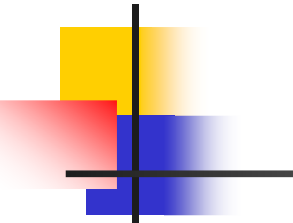


$$\begin{aligned}
 \mathbf{A} \mathbf{X} \mathbf{B} &= \begin{bmatrix} \sum_{j=1}^n a_{1j} \mathbf{x}_j^T \mathbf{B} \\ \vdots \\ \sum_{j=1}^n a_{mj} \mathbf{x}_j^T \mathbf{B} \end{bmatrix} = \begin{bmatrix} \left(\sum_{j=1}^n a_{1j} \mathbf{B}^T \mathbf{x}_j \right)^T \\ \vdots \\ \left(\sum_{j=1}^n a_{mj} \mathbf{B}^T \mathbf{x}_j \right)^T \end{bmatrix}
 \end{aligned}$$

The diagram illustrates the matrix multiplication $\mathbf{A} \mathbf{X} \mathbf{B}$ using block matrices. The matrix $\mathbf{A}$ is represented by a column vector of elements a_{ij} . The matrix $\mathbf{X}$ is represented by a column vector of vectors $\mathbf{x}_j^T$. The matrix $\mathbf{B}$ is represented by a column vector of vectors $\mathbf{B}^T \mathbf{x}_j$. The result is shown as a column vector of vectors $\left(\sum_{j=1}^n a_{ij} \mathbf{B}^T \mathbf{x}_j \right)^T$. The diagram uses color coding: blue for a_{ij} , red for $\mathbf{x}_j$, and black for $\mathbf{B}$. The summation is over $j=1$ to n . The matrix $\mathbf{A}$ is shown as a column vector of elements a_{1j} to a_{mj} . The matrix $\mathbf{X}$ is shown as a column vector of vectors $\mathbf{x}_j^T$. The matrix $\mathbf{B}$ is shown as a column vector of vectors $\mathbf{B}^T \mathbf{x}_j$. The result is shown as a column vector of vectors $\left(\sum_{j=1}^n a_{ij} \mathbf{B}^T \mathbf{x}_j \right)^T$. The diagram uses color coding: blue for a_{ij} , red for $\mathbf{x}_j$, and black for $\mathbf{B}$. The summation is over $j=1$ to n .



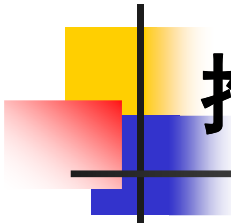
所以 $\text{vec}(\mathbf{AXB}) = \begin{bmatrix} \sum_{j=1}^n a_{1j} \mathbf{B}^T \mathbf{x}_j \\ \vdots \\ \sum_{j=1}^n a_{ij} \mathbf{B}^T \mathbf{x}_j \\ \vdots \\ \sum_{j=1}^n a_{mj} \mathbf{B}^T \mathbf{x}_j \end{bmatrix}$



$$= \begin{bmatrix} a_{11} \mathbf{B}^T & \cdots & a_{1j} \mathbf{B}^T & \cdots & a_{1n} \mathbf{B}^T \\ \vdots & & \vdots & & \vdots \\ a_{i1} \mathbf{B}^T & \cdots & a_{ij} \mathbf{B}^T & \cdots & a_{in} \mathbf{B}^T \\ \vdots & & \vdots & & \vdots \\ a_{m1} \mathbf{B}^T & \cdots & a_{mj} \mathbf{B}^T & \cdots & a_{mn} \mathbf{B}^T \end{bmatrix} \begin{bmatrix} \mathbf{x}_1 \\ \vdots \\ \mathbf{x}_j \\ \vdots \\ \mathbf{x}_n \end{bmatrix}$$

$$= (\mathbf{A} \otimes \mathbf{B}^T) \text{vec } \mathbf{X}$$





推论 设 $A \in \mathbf{C}^{m \times m}$, $B \in \mathbf{C}^{n \times n}$, $X \in \mathbf{C}^{m \times n}$, 则

$$\textcircled{1}. \text{vec}(AX) = (A \otimes I_n) \text{vec} X \quad (B = I_n)$$

$$\textcircled{2}. \text{vec}(XB) = (I_m \otimes B^T) \text{vec} X \quad (A = I_m)$$

$$\textcircled{3}. \text{vec}(AX + XB) = (A \otimes I_n + I_m \otimes B^T) \text{vec} X$$

定义 8.4 设 $A \in C^{m \times m}$, $B \in C^{n \times n}$, $X \in C^{m \times n}$, $F \in C^{m \times n}$, 则

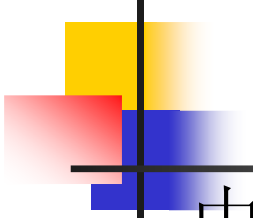
$$AX + XB = F \quad (8.21)$$

称为Lyapunov矩阵方程.

将矩阵方程(8.21)两端矩阵按行拉直,
由式(8.20)可得

$$(A \otimes I_n + I_m \otimes B^T) \text{vec } X = \text{vec } F \quad (8.22)$$

因矩阵方程(8.21)与线性方程组(8.22)等价,
故矩阵方程的求解问题转化为线性方程组
的求解问题.



由非齐次线性方程组的Kronecker定理知道
非齐次线性方程组有解的充要条件是：

系数矩阵的秩等于增广矩阵的秩。

因此Lyapunov矩阵方程(8.21)有解的充要条件是：

$$\text{rank}(A \otimes I_n + I_m \otimes B^T) = \text{rank}(A \otimes I_n + I_m \otimes B^T : \text{vec} F)$$

定义 8.5 设 $A \in C^{m \times m}$, $B \in C^{n \times n}$, 记

$$A \oplus B \triangleq A \otimes I_n + I_m \otimes B$$

称为矩阵 A 与 B 的 **Kronecker 和**.

有了矩阵 **Kronecker 和** 的定义, 则式(8.23)可归结为

定理 8.8 Lyapunov 矩阵方程

$$AX + XB = F$$

有解的充要条件是:

$$\text{rank}(A \oplus B^T) = \text{rank}(A \oplus B^T : \text{vec } F)$$



推论 若 $\det(A \oplus B^T) \neq 0$, 则 **Lyapunov**
矩阵方程 (8.21) 有 **惟一解**.

由 **定理** 8.2 上述 **结论** 可叙述为:
若 A, B **无互为反号的特征值**,
则方程 (8.21) 有 **惟一解**.

例 8.4 求解 Lyapunov 矩阵方程 $AX + XB = F$, 其中

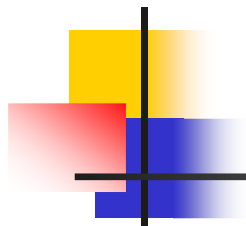
$$A = \begin{bmatrix} 1 & -1 \\ 0 & -1 \end{bmatrix}, \quad B = \begin{bmatrix} 0 & 2 \\ 1 & 0 \end{bmatrix}, \quad F = \begin{bmatrix} 1 & 3 \\ -2 & 2 \end{bmatrix}.$$

解 易见 A 的特征值为 $\lambda_1 = 1, \lambda_2 = -1$;

容易算得 B 的特征值为 $\mu_1 = \sqrt{2}, \mu_2 = -\sqrt{2}$.

由于 A, B 无互为反号的特征值,

故 Lyapunov 矩阵方程有惟一解.



$$\begin{aligned} \mathbf{A} \otimes \mathbf{I}_2 + \mathbf{I}_2 \otimes \mathbf{B}^T &= \begin{bmatrix} \boxed{1} & \boxed{0} & \boxed{-1} & \boxed{0} \\ \boxed{0} & \boxed{1} & \boxed{0} & \boxed{-1} \\ 0 & 0 & \boxed{-1} & \boxed{0} \\ 0 & 0 & \boxed{0} & \boxed{-1} \end{bmatrix} + \begin{bmatrix} \boxed{0} & \boxed{1} & 0 & 0 \\ \boxed{2} & \boxed{0} & 0 & 0 \\ 0 & 0 & \boxed{0} & \boxed{1} \\ 0 & 0 & \boxed{2} & \boxed{0} \end{bmatrix} \\ &= \begin{bmatrix} 1 & 1 & -1 & 0 \\ 2 & 1 & 0 & -1 \\ 0 & 0 & -1 & 1 \\ 0 & 0 & 2 & -1 \end{bmatrix} \end{aligned}$$



设 $X = \begin{bmatrix} x_1 & x_2 \\ x_3 & x_4 \end{bmatrix}$

则由式(8.22)有

$$\begin{bmatrix} 1 & 1 & -1 & 0 \\ 2 & 1 & 0 & -1 \\ 0 & 0 & -1 & 1 \\ 0 & 0 & 2 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \\ -2 \\ 2 \end{bmatrix}$$

解之 $x_1 = 0, x_2 = 1, x_3 = 0, x_4 = -2$.

于是 $X = \begin{bmatrix} 0 & 1 \\ 0 & -2 \end{bmatrix}$ 为 **Lyapunov** 矩阵方程的 **惟一解**. $\square$

定理8.9 如果 A, B 为稳定矩阵,即 A, B 的所有特征值都具有负实部,则Lyapunov矩阵方程

$$AX + XB = F$$

有惟一解

$$X(t) = - \int_0^{+\infty} e^{At} F e^{Bt} dt.$$

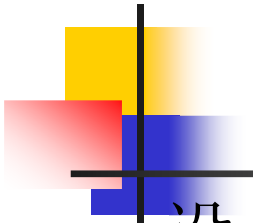
定理8.9 如果 A, B 为稳定矩阵,即 A, B 的所有特征值都具有负实部,则Lyapunov矩阵方程

$$AX + XB = F$$

有惟一解

$$X(t) = - \int_0^{+\infty} e^{At} F e^{Bt} dt.$$

证略


$$\sum_{k=1}^t A_k X B_k = F \text{ 型矩阵方程}$$

设 $A_k \in C^{m \times m}$, $B_k \in C^{n \times n}$, $X \in C^{m \times n}$, $F \in C^{m \times n}$, 求解矩阵方程

$$\sum_{k=1}^t A_k X B_k = F \quad (8.25)$$

将式(8.25)两端矩阵按行拉直有

$$\left[\sum_{k=1}^t (A_k \otimes B_k^T) \right] \text{vec } X = \text{vec } F \quad (8.26)$$

由于矩阵方程(8.25)与线性方程组(8.26)等价,
故与式(8.23)类似,可讨论方程(8.25)是否有解,
若有解是否惟一.

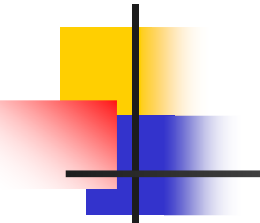


例 8.5 求解矩阵方程 $A_1 \mathbf{X} B_1 + A_2 \mathbf{X} B_2 = \mathbf{F}$, 其中

$$A_1 = \begin{bmatrix} 2 & 2 \\ 2 & -1 \end{bmatrix}, A_2 = \begin{bmatrix} 0 & 1 \\ -2 & -1 \end{bmatrix},$$

$$B_1 = \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix}, B_2 = \begin{bmatrix} 0 & 2 \\ -1 & 3 \end{bmatrix}, \mathbf{F} = \begin{bmatrix} 0 & -2 \\ 3 & -2 \end{bmatrix}.$$

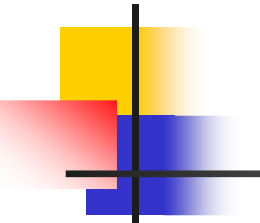
解 $B_1^T = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}, B_2^T = \begin{bmatrix} 0 & -1 \\ 2 & 3 \end{bmatrix}$, 设 $\mathbf{X} = \begin{bmatrix} x_1 & x_2 \\ x_3 & x_4 \end{bmatrix}$



$$\mathbf{A}_1 \otimes \mathbf{B}_1^T = \begin{bmatrix} 2\mathbf{B}_1^T & 2\mathbf{B}_1^T \\ 2\mathbf{B}_1^T & -1\mathbf{B}_1^T \end{bmatrix} = \begin{bmatrix} 2 & -2 & 2 & -2 \\ 0 & 2 & 0 & 2 \\ 2 & -2 & -1 & 1 \\ 0 & 2 & 0 & -1 \end{bmatrix},$$

$$\mathbf{A}_2 \otimes \mathbf{B}_2^T = \begin{bmatrix} 0\mathbf{B}_2^T & 1\mathbf{B}_2^T \\ -2\mathbf{B}_2^T & -1\mathbf{B}_2^T \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & -1 \\ 0 & 0 & 2 & 3 \\ 0 & 2 & 0 & 1 \\ -4 & -6 & -2 & -3 \end{bmatrix}.$$

由式(8.26)有

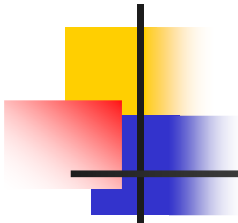

$$\begin{bmatrix} 2 & -2 & 2 & -3 \\ 0 & 2 & 2 & 5 \\ 2 & 0 & -1 & 2 \\ -4 & -4 & -2 & -4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 0 \\ -2 \\ 3 \\ -2 \end{bmatrix}.$$

解之该线性方程组有惟一解为

$$x_1 = 1, x_2 = 0, x_3 = -1, x_4 = 0.$$

故原矩阵方程有惟一解为

$$X = \begin{bmatrix} 1 & 0 \\ -1 & 0 \end{bmatrix}.$$



§ 8.4 矩阵微分方程

引理1 设 $A \in C^{m \times m}$, $B \in C^{n \times n}$, 则

$$(1) \quad e^{A \otimes I_n} = e^A \otimes I_n$$

$$(2) \quad e^{I_m \otimes B} = I_m \otimes e^B. \quad (8.27)$$



证 由**矩阵幂级数**的定义及**定理 8.1**之(2)可得

$$\begin{aligned} e^{A \otimes I_n} &= \sum_{k=0}^{\infty} \frac{1}{k!} (A \otimes I_n)^k = \sum_{k=0}^{\infty} \frac{1}{k!} (A^k \otimes I_n^k) \\ &= \sum_{k=0}^{\infty} \left(\frac{A^k}{k!} \otimes I_n \right) = \left(\sum_{k=0}^{\infty} \frac{A^k}{k!} \right) \otimes I_n = e^A \otimes I_n. \end{aligned}$$

故(1)成立. 同理可证(2). ■

引理2 设 $A \in C^{m \times m}$, $B \in C^{n \times n}$, 则


$$e^{(A \otimes I_n + I_m \otimes B)} = e^A \otimes e^B = e^B \otimes e^A \quad (8.28)$$

证 因为

$$(A \otimes I_n)(I_m \otimes B) = A \otimes B = (I_m \otimes B)(A \otimes I_n)$$

这说明 $(A \otimes I_n)$ 与 $(I_m \otimes B)$ 可交换.

由 **定理 5.8** 之 (1) 及 **引理 1** 可得

$$\begin{aligned} e^{(A \otimes I_n + I_m \otimes B)} &= e^{A \otimes I_n} \cdot e^{I_m \otimes B} \\ &= (e^A \otimes I_n)(I_m \otimes e^B) = e^A \otimes e^B \end{aligned}$$

同理可证另一等式成立. ■



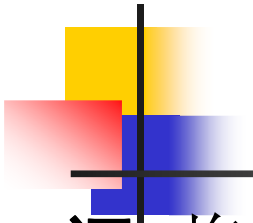
定理 8.10 矩阵微分方程

$$\begin{cases} \frac{d\mathbf{X}(t)}{dt} = \mathbf{A}\mathbf{X}(t) + \mathbf{X}(t)\mathbf{B} \\ \mathbf{X}(0) = \mathbf{X}_0 \end{cases} \quad (8.29)$$

的解为

$$\mathbf{X}(t) = e^{\mathbf{A}t} \mathbf{X}_0 e^{\mathbf{B}t} \quad (8.30)$$

其中 $\mathbf{A} \in \mathbb{C}^{m \times m}$, $\mathbf{X} \in \mathbb{C}^{m \times n}$, $\mathbf{B} \in \mathbb{C}^{n \times n}$.



证 将**矩阵微分方程** (8.29) 两端矩阵**按行拉直**有

$$\begin{cases} \frac{d}{dt}(\text{vec } X(t)) = (A \otimes I_n + I_m \otimes B^T) \text{vec } X(t) \\ \text{vec } X(t) = \text{vec } X_0 \end{cases} \quad (8.31)$$

于是问题归结为求解**常系数齐次线性微分方程组初值问题**(8.31).

由**定理 5.10**中式(5.39)及**引理 2, 定理 8.7**可得

$$\text{vec } X(t) = e^{(A \otimes I_n + I_m \otimes B^T)t} \cdot \text{vec } X_0$$

$$= (e^{At} \otimes e^{B^T t}) \text{vec } X_0 = \text{vec}(e^{At} X_0 (e^{B^T t})^T)$$

而 $(e^{B^T t})^T = \left(\sum_{k=0}^{\infty} \frac{1}{k!} (B^T)^k t^k \right)^T = \sum_{k=0}^{\infty} \frac{1}{k!} B^k t^k = e^{Bt}$

故

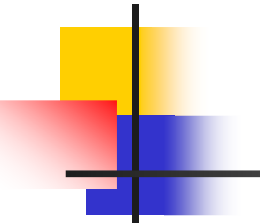
$$\text{vec } X(t) = \text{vec}(e^{At} X_0 e^{Bt}).$$

于是矩阵微分方程初值问题 (8.29) 的解为

$$X(t) = e^{At} X_0 e^{Bt}.$$



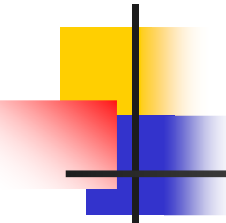
例 8.6 求解矩阵微分方程的初值问题


$$\begin{cases} \frac{d\mathbf{X}(t)}{dt} = \begin{bmatrix} -1 & 0 \\ 0 & 2 \end{bmatrix} \mathbf{X}(t) + \mathbf{X}(t) \begin{bmatrix} 1 & -1 \\ 0 & 2 \end{bmatrix} \\ \mathbf{X}(0) = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \end{cases}$$

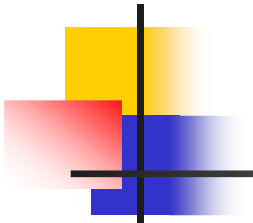
解 令 $\mathbf{A} = \begin{bmatrix} -1 & 0 \\ 0 & 2 \end{bmatrix}$, $\mathbf{B} = \begin{bmatrix} 1 & -1 \\ 0 & 2 \end{bmatrix}$,

因 $\mathbf{A}$ 是对角阵, 故 $e^{\mathbf{A}t} = \begin{bmatrix} e^{-t} & 0 \\ 0 & e^{2t} \end{bmatrix}$.

以下用**Laplace变换**的方法计算 e^{Bt} .


$$(sI - B)^{-1} = \begin{bmatrix} s-1 & 1 \\ 0 & s-2 \end{bmatrix}^{-1}$$
$$= \frac{1}{(s-1)(s-2)} \begin{bmatrix} s-2 & -1 \\ 0 & s-1 \end{bmatrix} = \begin{bmatrix} \frac{1}{s-1} & \frac{1}{s-1} - \frac{1}{s-2} \\ 0 & \frac{1}{s-2} \end{bmatrix}$$

$$e^{Bt} = L^{-1}[(sI - B)^{-1}] = \begin{bmatrix} e^t & e^t - e^{2t} \\ 0 & e^{2t} \end{bmatrix}$$



由定理 8.9 知

$$X(t) = e^{At} X(0) e^{Bt}$$

$$= \begin{bmatrix} e^{-t} & 0 \\ 0 & e^{2t} \end{bmatrix} \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} e^t & e^t - e^{2t} \\ 0 & e^{2t} \end{bmatrix}$$

$$= \begin{bmatrix} -1 & -1 + e^t \\ 0 & e^{5t} \end{bmatrix}.$$





例 8.7 若矩阵方程 $AXB = D$ 不相容, 则它的
极小范数最小二乘解 $X_0 = A^+DB^+$.

证 将矩阵方程两端拉直有 $\text{vec}(AXB) = \text{vec } D$

矩阵方程 $AXB = D \Leftrightarrow$ 线性方程组 $\text{vec}(AXB) = \text{vec } D$.

因为矩阵方程 $AXB = D$ 不相容, 故线性方程组

$$(A \otimes B^T) \text{vec } X = \text{vec } D$$

不相容.

其极小范数最小二乘解为

$$\begin{aligned} \text{vec } X_0 &= (A \otimes B^T)^+ \text{vec } D \\ &= [A^+ \otimes (B^T)^+] \text{vec } D \\ &= [A^+ \otimes (B^+)^T] \text{vec } D \\ &= \text{vec}(A^+ D B^+) \end{aligned}$$

这说明 与之对应地 $X_0 = A^+ D B^+$ 正是不相容的
矩阵方程 $AXB = D$ 的极小范数最小二乘解. ■